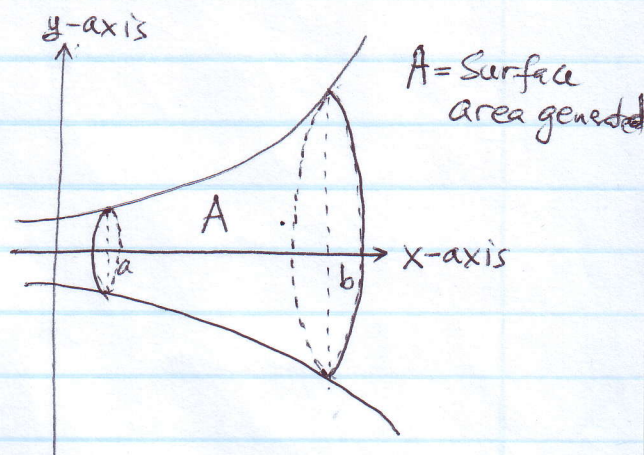
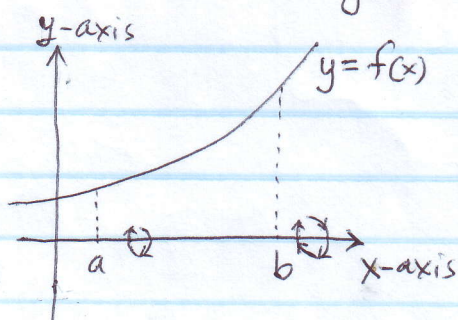


3) SURFACE OF REVOLUTION

If the curve $y = f(x)$ is rotated about an axis, a surface area A will be generated.



The surface area generated:

Remember this *

$$A = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

if $y = f(x)$ is Cartesian Curve

Remember this **

$$A = 2\pi \int_{\theta_1}^{\theta_2} y \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

if $\begin{cases} x = g(\theta) \\ y = h(\theta) \end{cases}$ is a parametric Curve

Example 1

Find the area generated when the arc of the parabola $y^2 = 8x$ between $x = 0$ and $x = 2$, rotates abt the x-axis.

Solution

$$A = 2\pi \int_0^2 y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

$$y^2 = 8x \rightarrow y = \sqrt{8x} = 2\sqrt{2}\sqrt{x}$$

$$\frac{dy}{dx} = 2\sqrt{2} \cdot \frac{1}{2\sqrt{x}} = \frac{\sqrt{2}}{\sqrt{x}}$$

$$\text{Then } 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{2}{x} = \frac{x+2}{x}$$

$$\therefore \left(\frac{dy}{dx}\right)^2 = \frac{2}{x}$$

$$A = 2\pi \int_0^2 2\sqrt{2}\sqrt{x} \sqrt{\frac{x+2}{x}} dx$$

$$A = 2\pi \int_0^2 2\sqrt{2} \sqrt{x+2} dx$$

$$A = 4\sqrt{2} \pi \int_0^2 (x+2)^{1/2} dx = 4\sqrt{2} \pi \cdot \frac{2}{3} (x+2)^{3/2} \Big|_0^2$$

$$= \frac{8\sqrt{2}}{3} \pi [4^{3/2} - 2^{3/2}]$$

$$\therefore A = 61.3 \text{ unit}^2$$

Example 2

Find the area generated when the curve

$$x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta)$$

between $\theta = 0$ and $\theta = \pi$, rotates about the x-axis.

Solution

$$A = 2\pi \int_0^\pi y \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$\left. \begin{aligned} \frac{dx}{d\theta} &= a(1 - \cos \theta) \\ \frac{dy}{d\theta} &= a \sin \theta \end{aligned} \right\} \quad \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta$$

$$= a^2(1 + \cos^2 \theta - 2 \cos \theta) + a^2 \sin^2 \theta$$

$$= a^2 + a^2(\cos^2 \theta + \sin^2 \theta) - 2a^2 \cos \theta$$

$$= 2a^2(1 - \cos \theta)$$

$$= 4a^2 \sin^2 \frac{\theta}{2}, \quad \because 1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$$

Then:

$$y \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}$$

$$= a(1 - \cos \theta) \left(2a \sin \frac{\theta}{2}\right)$$

$$= a \cdot \left(2 \sin^2 \frac{\theta}{2}\right) \left(2a \sin \frac{\theta}{2}\right)$$

$$= 4a^2 \left(1 - \cos^2 \frac{\theta}{2}\right) \sin \frac{\theta}{2}$$

$$= 4a^2 \left(\sin \frac{\theta}{2} - \cos^2 \frac{\theta}{2} \sin \frac{\theta}{2}\right)$$

Surface Area:

$$A_s = 8a^2 \pi \int_0^\pi \left[\sin \frac{\theta}{2} - \cos^2 \frac{\theta}{2} \sin \frac{\theta}{2}\right] d\theta$$

$$A = 8a^2\pi \left[-2 \cos \frac{\theta}{2} \Big|_0^\pi + \frac{2}{3} \cos^3 \frac{\theta}{2} \Big|_0^\pi \right]$$

$$= 8a^2\pi \left[-2(0-1) + \frac{2}{3}(0-1) \right]$$

$$= \frac{32a^2\pi}{3} \text{ units}^2$$
